

Rugved M. Zarkar

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EDUCATION

University of Maryland, College Park – College Park, MD

Expected December 2027

Bachelor of Science in Computer Science

GPA: 3.83/4.00

Relevant Coursework: Advanced Data Structures, Machine Learning, Compilers, Computer Vision, Operating Systems, Database Systems, Algorithms, Object-Oriented Programming

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C, C++, HTML/CSS, MATLAB

Frameworks: React.js, FastAPI, PyTorch, TensorFlow, Tailwind, LangChain, HuggingFace, Scikit-Learn, MySQL

Developer Tools: Amazon Web Services, Git, GitHub, Linux, Docker, Supabase, REST APIs, Node.js, Bash

Certifications: AWS Certified Cloud Practitioner, AWS Solutions Architect, Autodesk Fusion Certified

PROFESSIONAL EXPERIENCE

Amazon Web Services

August 2026

Incoming Software Engineer Intern

Arlington, VA

- Incoming intern on the Domain Name Services team

IBM

May 2026 - Present

Software Engineer Intern

Austin, TX

- Architected and optimized a recursive DNS platform using DNSDist, Unbound, and SDNLB, scaling throughput **10×** from **100K** to **1M+ QPS** with **sub-1% loss** across 30 clients, 2 DNSDist nodes, and 3 Unbound nodes
- Increased DNSDist throughput **19%** through multi-instance CPU pinning and end-to-end DNSDist, Unbound, Linux kernel, systemd, and network tuning
- Built Python metrics exporters, customized Grafana/Osprey dashboards, and developed **Ansible automation** that cut node provisioning from days to **under 3 minutes** for IBM's Classic-to-VPC migration

University of Maryland Institute for Advanced Computer Studies

Aug 2024 – Present

Software Research Assistant

College Park, MD

- Engineered C++ and Python firmware for an **ESP32-S3** wearable robot with a touchscreen UI displaying anatomical imagery, integrating an IMU sensor pipeline, and WiFi communication stack
- Extended the **ACM IMWUT**-published Calico platform to support educational programming environments; project featured in **IEEE Spectrum** and The Verge, and demonstrated at **CHI** and **UbiComp**

RAGent LLC

Jan 2025 – Apr 2026

Founder & Lead Software Developer

Ellicott City, MD

- Founded an AI startup generating **\$20K+ ARR** across a **20+** employee organization; architected RAG pipelines using llama-index, FastAPI, OpenAI API, and Weaviate for hybrid vector + keyword search and parallel document retrieval across dynamic enterprise and education deployments
- Delivered an AI workflow for **sustainED (McKinsey-backed)** on AWS-native infrastructure (Terraform IAC, Aurora serverless v2, Bedrock Titan + claude 3.5 Haiku) that translates/simplifies **7,000+ restricted research documents**, reducing review time by **35%** and improving accessibility for Southern African education ministries

National Institute of Standards and Technology

May 2025 – Mar 2026

Software Engineer Intern

Gaithersburg, MD

- Engineered AI-driven metrology pipelines using **PyTorch**, TensorFlow, and Scikit-Learn to process **1,000+ SEM images**, including a fine-tuned **DDPM** for image enhancement, reducing scan points by **40%**, improving fidelity by **20%**, and lowering measurement uncertainty by **50%**

PROJECTS

HarnessFlow - AI Agent Harness

May 2026

- Built an open-source framework that lets developers define **AI** agent pipelines in **YAML** and compiles them into deterministic, **production-reliable workflows**; solving the issue of unpredictable behavior in production systems
- Compiled YAML workflows into Temporal-orchestrated runs executed by a **polyglot Go** (Connect-Go, sqlc) and Python runtime with model fallback and approval-gate signals; instrumented with **OpenTelemetry GenAI** and an eval-gated CI using **LLM-as-judge**, embedding similarity, and cost/latency scorers